Alexander W. Kohn

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Summary

Data scientist with a Ph.D. in physical chemistry and eight years of production ML in security and trust & safety: malware detection and behavioral authentication at Cylance, then detection of toxic chat, offensive names, and cheat devices at Blizzard. My work centers on careful evaluation, clean labels, and models that hold up in production. Since 2023 it has also included LLM labeling, fine-tuning, and retrieval-based assistants.

Experience

Blizzard Entertainment **May 2021 – Present**
Player Interactions and Trust *Irvine, CA*
Senior Data Scientist **2022 – Present**
Data Scientist **2021 – 2022**

- Own production text classifiers that screen player names and in-game chat for harmful content in games with **millions of daily players**, covering data collection, training, evaluation, deployment, and retraining.
- Built and maintain the player naming model across four major versions; the current model reaches **95% recall at 99% precision**. Worked with policy and operations teams over several rounds to refine the harm categories and acceptable-name lists, including how protected-class, religious, and LGBT terms are handled.
- Raised toxicity and hate-speech classification accuracy by **about 3%** and added support for more languages, building evaluation datasets and annotation workflows for each new language.
- Gained another **2%** in chat model accuracy from knowledge distillation and class weighting, including training several teacher models and domain-pretrained BERT encoders.
- Cleaned up training labels by using model ensembles to find mislabeled examples, testing label-noise mitigation methods, and analyzing prediction flips between model versions.
- Built a nine-category chat classifier covering two tiers each of toxicity, hate speech, and prosocial behavior, plus real-money-trading spam, service spam, and phishing. Wrote a policy definition for each category, had an LLM label real chat against them, and fine-tuned Llama 3.1 8B with LoRA. Sampled training chat to cover heavily reported messages, strings earlier models wrongly flagged, and eight non-Latin scripts. Analysts use it in their review workflows, and it is being set up to act on phishing and spam.
- Improved the training pipeline for a production detector of hardware cheat devices, which pass mouse-and-keyboard or scripted input off as controller input. Made training reproducible between dev and prod and raised recall by **about 4 percentage points** while also improving precision.
- Modeled each cheat reporter's reliability as the share of their reports later confirmed by bans (Beta-Binomial with an empirical-Bayes prior, adjusted for ban delay), then combined reporters' evidence into a per-player score for ranking suspects.
- Built the **company's first internal LLM chat agents**: a retrieval-augmented assistant for game lore and quest content, and a customer-support assistant that suggests replies, summarizes and categorizes tickets, and translates messages. Took both from prototype to production on LangChain, vector databases, Flask, Docker, Postgres, and LangSmith, with scheduled GDPR data purges.
- Used double machine learning to estimate how exposure to cheating and toxicity affects player engagement and retention, adding confounders over several rounds of analysis.

BlackBerry Cylance **Aug 2018 – Apr 2021**
Data Scientist, Research *Irvine, CA*

- One of four developers of *Persona*, an unsupervised system that continuously authenticates users from their behavior. Wrote the Bayesian conjugate models for process and registry activity and built the evaluation pipeline and metrics. Co-inventor on the resulting patent.
- Built LightGBM models on NLP features that classify malicious PHP with **97.1% accuracy**.
- Researched and tested defenses against adversarial string-based attacks on a production malware classifier.
- Improved training labels for executable files by combining noisy verdicts from multiple antivirus vendors into one smoothed label.
- Built ingestion for heterogeneous event-sequence data from an endpoint security product.

Massachusetts Institute of Technology **Nov 2013 – Jan 2018**
Graduate Researcher, Van Voorhis Lab *Cambridge, MA*

- Developed excited-state electronic structure models of non-radiative decay in organic electronics.
- Used a **graph neural network** to predict optimal tuning parameters for molecules in the Harvard Clean Energy Project database.
- Wrote Python pipelines for electronic structure calculations and sped up CDFT-CI analytic gradients.

Education

Massachusetts Institute of Technology, Ph.D., Physical Chemistry **Jan 2018**
Harvey Mudd College, B.S., Chemistry, *summa cum laude* **May 2013**

Technical Skills

Languages	Python, SQL, Bash
ML / NLP	PyTorch, Hugging Face, scikit-learn, LightGBM, distillation, LoRA fine-tuning, LLM evaluation
LLM Systems	LangChain, RAG, vector databases, prompt optimization, LangSmith, Weights & Biases
Inference	Causal inference (double ML), Bayesian modeling, survival analysis, experiment design
Platform	Databricks, Spark, dbt, Postgres, Docker, Git, CI/CD

Patent

- M. T. Wojnowicz, D. H. Nguyen, **A. W. Kohn**. Bayesian Continuous User Authentication. U.S. Patent 11,544,358 B2, granted Jan. 2023. Assigned to Cylance Inc.

Selected Publications

- **A. W. Kohn**, Z. Lin, T. Van Voorhis. Toward Prediction of Nonradiative Decay Pathways in Organic Compounds I: The Case of Naphthalene Quantum Yields. *J. Phys. Chem. C*, 2019.
- Z. Lin, **A. W. Kohn**, T. Van Voorhis. Toward Prediction of Nonradiative Decay Pathways in Organic Compounds II: Two Internal Conversion Channels in BODIPYs. *J. Phys. Chem. C*, 2020.
- **A. W. Kohn**, D. McMahon, S. Wen, T. Van Voorhis. The Impact of Carrier Delocalization and Interfacial Electric Field Fluctuations on Organic Photovoltaics. *J. Phys. Chem. C*, 2017.